

closure_evidence_20260925

BOB-143 closure evidence — 2026-09-25

Item: BOB-143 — Orphaned `.worktrees/` dirs (46M, unresolvable gitdir) pollute gate scan scope and manufacture false BOB-126-class findings.

Fix implemented: option (b) from the item's own text — add `.worktrees/` to the gate scan-scope exclusion list as a §11.4.224(E)-fenced, checked-in, justified entry. **Deletion was NOT performed** (and, as established below, was not even applicable in this checkout — see “Pre-fix investigation”). Nothing inside `.worktrees/` was touched.

0. Pre-fix investigation (root cause first, per systematic-debugging)

Before writing anything, I verified the item's premise against the live checkout, per the constitution's “reproduce first” / “no guessing” discipline.

```
$ test -d .worktrees && echo "EXISTS: .worktrees" || echo  
"MISSING: .worktrees"  
MISSING: .worktrees
```

```
$ find . -maxdepth 4 -iname "*worktree*" -not -path "./submodules/  
*" -not -path "./.git/*"  
./docs/proposals/subagent-worktree-isolation.md  
./docs/proposals/subagent-worktree-isolation.html  
./docs/proposals/subagent-worktree-isolation.docx  
./docs/proposals/subagent-worktree-isolation.pdf  
./constitution/scripts/multitrack/multitrack_resolve_worktree.sh
```

Finding: `.worktrees/ci-split-workflows/` and `.worktrees/completion-initiative-phase-0/` do **not exist** in this checkout today. They were local, gitignored, host-side scratch directories at the time BOB-143 was filed (2026-08-20); by 2026-09-25 they are simply absent from disk (this session's checkout never had them, or they were cleaned up by unrelated host activity in the interim — either way, git carries no record of their removal since they were never tracked). This does not change the task: the item's own acceptance criterion explicitly names exclusion as a valid, sufficient close path *independent of whether the directories currently exist* (“or `.worktrees/` is added to a checked-in ... exclusion list”), and the constitution's own §11.4.6 boundary applies — I did not invent or assume their presence, I measured it and proceeded on what the checkout actually shows. A synthetic, deterministic fixture (mirroring the precedent `tests/pre_build/test_check_cm_go_toolchain_matches_builder.sh` / BOB-194 pattern, whose own comment states “does not depend on a worktree happening to exist on disk, which is how the real-tree case caught it only by luck”) is therefore the correct RED/GREEN vehicle here, not a live `.worktrees/tree`.

1. Investigation of the two measured findings

1a. `cm_test_mock_pid_explicit_int` (CM-TEST-MOCK-PID-EXPLICIT-INT)

Traced the two invocation paths of this gate in the repository:

- `scripts/pre_build/check_cm_test_mock_pid_explicit_int.sh` — the thin wrapper `pre_build_verification.sh` invariant 28 uses. Its `DEFAULT_SCAN_ROOTS=("tests")` — it only ever walks `$REPO_ROOT/tests`, so a top-level `.worktrees/` directory is structurally outside its scan and was never the source of the BOB-143 finding.
- `constitution/scripts/gates/cm_test_mock_pid_explicit_int.sh` (the shared engine) invoked **directly** by `scripts/verify-all-constitution-rules.sh` (boba's §11.4.32 sweep — the exact instrument BOB-143 cites: “Measured 2026-08-20 by the §11.4.32 sweep”) with `--root @ROOT@` (the whole repository) via `config/constitution-sweep.conf`'s `DEFAULT` row. **This** is where the false findings were sourced.

Reading scripts/verify-all-constitution-rules.sh, this half is **already fixed** — by commit c0ea01c (title unrelated to BOB-143, but its body lands the fix), which added:

```
_BOB143_SWEEP_EXCLUDE=".git node_modules vendor .venv __pycache__
scripts/gates out build dist .worktrees"
export DANGEROUS_COMBO_EXCLUDE="${DANGEROUS_COMBO_EXCLUDE:-
$_BOB143_SWEEP_EXCLUDE}"
...
_BOB152_VENDORED_EXCLUDE="${_BOB143_SWEEP_EXCLUDE} submodules"
export MOCK_PID_GUARD_EXCLUDE="${MOCK_PID_GUARD_EXCLUDE:-
$_BOB152_VENDORED_EXCLUDE}"
export ORACLE_GUARD_EXCLUDE="${ORACLE_GUARD_EXCLUDE:-
$_BOB152_VENDORED_EXCLUDE}"
export KILLPG_GUARD_EXCLUDE="${KILLPG_GUARD_EXCLUDE:-
$_BOB152_VENDORED_EXCLUDE}"
```

.worktrees is present in _BOB143_SWEEP_EXCLUDE, which every one of these four env-var exports inherits. Item BOB-143 was never closed to reflect this, so I treated this as an **already-fixed half needing a regression guard**, not something to re-implement. Confirmed my mutation test above (_BOB143_SWEEP_EXCLUDE no longer names .worktrees abort) catches a future regression of exactly this wiring.

1b. “6 of 57 missing anchor carrier” propagation-gate findings

Traced to scripts/verify-all-constitution-rules.sh’s delegation of the whole CM-COVENANT-114-<N>-PROPAGATION family to constitution/scripts/gates/covenant_propagation_suite.sh gates --root "\$root" (boba’s own root). The suite pre-computes a carrier list with its own find, and the shared engine (constitution/scripts/gates/lib/covenant_propagation_engine.sh) does its own carrier discovery too when not given a precomputed list — **neither pruned .worktrees**, and **neither may hardcode a project path** (both are upstream, inherited by reference per §11.4.28/§11.4.177 — a boba-specific literal must never be injected into them). Both already read a **consumer-owned, checked-in exclusion file**:

```
default: <root>/config/covenant_propagation_exclusions.tsv
override: $COVENANT_PROPAGATION_EXCLUSIONS
```

documented in the engine’s own header as “the §11.4.224(E) exclusion fence + the §11.4.135 checked-in-exemption-map pattern”, with a **closed class set** {vendored-third-party | generated-code | non-shipping-fixtures | filename-collision} the engine validates and fails closed (exit 2 / BLIND) on any row outside it or with no

justification. **This file did not exist in boba** (`config/covenant_propagation_exclusions.tsv` was absent). This is the genuinely unaddressed half of BOB-143, and is the actual fix delivered here.

I found a real-world precedent of this exact mechanism already in production use in a sibling project (`/home/milosvasic/Projects/vasic/config/ covenant_propagation_exclusions.tsv`, read-only, for format reference only — not part of this repository) — confirming the schema and that this is the established, intended consumer-side mechanism rather than something invented for this fix.

2. Files found and modified

File	Action	Why
<code>config/covenant_propagation_exclusions.tsv</code>	Created	The checked-in, §11.4.224(E)-fence exclusion-list entry BOB-143 optically asks for. Read by default by both <code>constitution/scripts/gates/lib/covenant_propagation_engine.sh</code> (single-gate carrier discovery) and <code>constitution/scripts/gates/covenant_propagation_suite.sh</code> (the batch runner's own carrier precompute) — no wiring change needed anywhere else, since both already default to <code>\${root}/config/covenant_propagation_exclusions</code> and <code>--root</code> for boba's own §11.4.3 sweep is always boba's repo root. §1.1 paired-mutation proof, mirrored the proven shape of <code>tests/pre_build/test_bob152_vendored_exclude_scope</code> (the closest precedent — same exclusion scope class, same sweep script). The new exclusion row is well-formed, load-bearing on BOTH the direct-engine invocation path and the production <code>covenant_propagation_suite.sh</code> invocation path, and regression-guards the already-fixed <code>MOCK_PID/ORA/KILLPG/DANGEROUS_COMBO</code> wiring.
<code>tests/pre_build/test_bob143_worktrees_exclude_scope.sh</code>	Created	

File	Action	Why
		scripts/verify-all-constitution rules.sh.

Files explicitly NOT touched (per scope): scripts/
pre_build_verification.sh (a concurrent, unrelated 135-line diff was
already present in the working tree when I started — not made by me,
confirmed by `git diff --stat` showing changes I never authored),
.gitignore, anything under .worktrees/ (doesn't exist), download-
proxy/, plugins/, docs/guides/, tests/ownership/, tests/integration/
test_container*, docs/workable_items.db, the workable-items CLI.

Why scripts/pre_build_verification.sh was NOT the fix location:
the constitution submodule's engine and suite explicitly forbid a
project literal being hardcoded into them (§11.4.28/§11.4.177 —
inherited by reference, never copied, never given project-specific
context), and both already implement a documented, purpose-built,
checked-in consumer-DATA exclusion mechanism reading config/
covenant_propagation_exclusions.tsv by default. Adding the row there
is the mechanism these files were designed for; editing
pre_build_verification.sh was neither necessary (its own CM-
COVENANT-PROPAGATION-SUITE invocation at “Invariant 32” passes
--root "\${PROJECT_ROOT}/constitution" — the constitution
submodule's root, not boba's — so it was never the source of the
.worktrees-sourced findings the item measured and needs no change
here) nor in scope (it is explicitly excluded by the task).

3. RED — before the fix

Manual reproduction against a synthetic fixture (since the real
.worktrees/ directories no longer exist — see §0), run BEFORE config/
covenant_propagation_exclusions.tsv existed:

```
$ mkdir -p "$WORK/.worktrees/probe" "$WORK/worktrees/control"
$ printf 'This is a stale orphaned worktree copy. No anchor block
here.\n' > "$WORK/.worktrees/probe/CLAUDE.md"
$ printf 'This is a stale sibling non-dot worktrees dir. No anchor
block here.\n' > "$WORK/worktrees/control/CLAUDE.md"
$ printf '**§11.4.230 – Parallelized-pipeline methodology.** Real
correct anchor block text so at least one carrier passes.\n' >
"$WORK/CLAUDE.md"

$ bash constitution/scripts/gates/
cm_covenant_114_230_propagation.sh --root "$WORK" --quiet
MISSING      worktrees/control/CLAUDE.md – zero 11.4.230 block-
```

```

starts (bare-literal citations elsewhere do not count,
§11.4.201(7)(a))
    MISSING      .worktrees/probe/CLAUDE.md  – zero 11.4.230 block-
starts (bare-literal citations elsewhere do not count,
§11.4.201(7)(a))
-----
CM-COVENANT-114-230-PROPAGATION: 1 single-block-PRESENT, 0
POINTER-INHERITANCE-SKIP, 2 MISSING/DUPLICATED, 0 DIVERGENT
(anchor 11.4.230) under $WORK
    CM-COVENANT-114-230-PROPAGATION: FAIL – anchor-block integrity
violated for §11.4.230
rc=1

```

This reproduces the item’s exact measured shape: a .worktrees/** carrier manufacturing a false MISSING finding indistinguishable from a real one (§11.4.201(1) false-positive refusal).

Formal paired-mutation proof, run with the fix FILE ABSENT (config/covenant_propagation_exclusions.tsv moved aside):

```

$ mv config/covenant_propagation_exclusions.tsv /tmp/...bak
$ bash tests/pre_build/test_bob143_worktrees_exclude_scope.sh
== BOB-143 .worktrees/ exclude scope proof ==
gates dir: /home/milosvasic/Projects/boba/constitution/scripts/
gates
exclusions file: /home/milosvasic/Projects/boba/config/
covenant_propagation_exclusions.tsv
ABORT: the fix (config/covenant_propagation_exclusions.tsv) is
absent – nothing to prove
EXIT=3
$ mv /tmp/...bak config/covenant_propagation_exclusions.tsv  #
restored

```

Second mutation — file present but the load-bearing row dropped (the more realistic “someone edited the file and lost the row” regression):

```

$ cp config/covenant_propagation_exclusions.tsv /tmp/...real
$ printf '# mutated: .worktrees row dropped\n*/some-other-
dir\tgenerated-code\tunrelated placeholder row\n' > config/
covenant_propagation_exclusions.tsv
$ bash tests/pre_build/test_bob143_worktrees_exclude_scope.sh
== BOB-143 .worktrees/ exclude scope proof ==
...
ABORT: expected exactly 1 '*/.worktrees' row in ../config/
covenant_propagation_exclusions.tsv, found 0
EXIT=3
$ cp /tmp/...real config/covenant_propagation_exclusions.tsv  #
restored, byte-identical (diff -q confirmed)

```

Third mutation — regression-guard for the *already-fixed* half (BOB-143's CM-TEST-MOCK-PID-EXPLICIT-INT wiring in the sweep script itself):

```
$ sed -i 's/... dist .worktrees"/... dist"/' scripts/verify-all-constitution-rules.sh
$ bash tests/pre_build/test_bob143_worktrees_exclude_scope.sh
== BOB-143 .worktrees/ exclude scope proof ==
row confirmed: */.worktrees filename-collision <justification, 1019 chars>
ABORT: _BOB143_SWEEP_EXCLUDE no longer names .worktrees (got: .git node_modules vendor .venv __pycache__ scripts/gates out build dist) – the already-measured-fixed half of BOB-143 regressed
EXIT=3
$ cp /tmp/...bak scripts/verify-all-constitution-rules.sh #
restored, diff -q confirmed identical to HEAD
```

All three mutations correctly fail the test (exit 3 — an honest ABORT that refuses to certify a verdict when the fix's own precondition is absent, matching the same convention tests/pre_build/test_bob152_vendored_exclude_scope.sh already uses for its own missing-precondition case).

4. GREEN — after the fix

```
$ bash tests/pre_build/test_bob143_worktrees_exclude_scope.sh
== BOB-143 .worktrees/ exclude scope proof ==
gates dir: /home/milosvasic/Projects/boba/constitution/scripts/gates
exclusions file: /home/milosvasic/Projects/boba/config/covenant_propagation_exclusions.tsv
row confirmed: */.worktrees filename-collision <justification, 1019 chars>
sweep wiring confirmed: _BOB143_SWEEP_EXCLUDE still
names .worktrees (feeds MOCK_PID/ORACLE/KILLPG/DANGEROUS_COMBO_EXCLUDE)

---- direct-engine: cm_covenant_114_230_propagation.sh ----
baseline (empty exclusions file): probe seen (1 hit(s)), control seen (1 hit(s)) – instrument PROVEN seeing both locations
PASS: with the real BOB-143 exclusion active, .worktrees/-rooted probe EXCLUDED (0 hits, and the ∅ EXCLUDED line was printed), worktrees/-rooted control STILL CAUGHT (1 hit(s)).

---- suite-path: covenant_propagation_suite.sh gates (production invocation shape) ----
baseline (empty exclusions file): anchor 230 row reports 2 MISSING/DUPLICATED (probe + control) – instrument PROVEN seeing both locations
```

PASS: via the suite path, with the real BOB-143 exclusion active anchor 230's row reports exactly 1 MISSING/DUPLICATED (the worktrees/ control only – the .worktrees/ probe's MISSING dropped out, i.e. was excluded from discovery).

VERDICT: PASS – BOB-143's .worktrees/ exclusion is load-bearing (excluded) while worktrees/ (no dot) is still caught, on both the direct-engine and the production suite invocation paths; the already-fixed MOCK_PID/ORACLE/KILLPG/DANGEROUS_COMBO wiring still names .worktrees.
EXIT=0

Manual confirmation against the real boba root (proves the fix is actually wired for real use, not just the hermetic fixture):

```
$ bash constitution/scripts/gates/  
cm_covenant_114_230_propagation.sh --root . --quiet | head -3  
CM-COVENANT-114-230-PROPAGATION: consumer exclusions applied (1)  
from /home/milosvasic/Projects/boba/config/  
covenant_propagation_exclusions.tsv:  
  o EXCLUDED */.worktrees [filename-collision]  
  MISSING    AGENTS.md – zero 11.4.230 block-starts ...
```

(The AGENTS.md / challenges/fixtures/anchor_block_integrity/** MISSING lines that follow are **pre-existing, unrelated** findings — AGENTS.md genuinely lacks the §11.4.230 block today, and challenges/fixtures/ anchor_block_integrity/** are the propagation engine's OWN golden-bad/ golden-good meta-test fixtures, deliberately missing blocks by design. Both are out of BOB-143's scope and untouched by this fix — confirming the exclusion did not widen to swallow anything beyond .worktrees.)

5. Proof the fix does NOT create a blind spot (§11.4.201(6))

This is built into the test itself (worktrees/ non-dot control, always proven still-caught in every scenario above) rather than a separate step: every PASS in §4 explicitly asserts the sibling worktrees/ bob143_control/ CLAUDE.md (no leading dot — same word, deliberately similar name, genuinely different path) is **still flagged MISSING** with the exclusion active. The exclusion glob is */.worktrees (an exact directory-name match via find -path), which cannot match a directory named worktrees (no leading dot) — proven empirically, not merely asserted, in both the direct-engine and the suite invocation paths.

6. No regression — existing tests re-run clean

```
$ bash -n tests/pre_build/*.sh # every file in the directory
syntax sweep done (0 failures)

$ timeout 120 bash constitution/scripts/gates/
cm_covenant_114_230_propagation_mutation_test.sh
...
    META PASS: CM-COVENANT-114-230-PROPAGATION is a genuine (non-
bluff) gate
EXIT=0

$ bash scripts/pre_build/check_cm_test_mock_pid_explicit_int.sh
    scope: default (tests), repo-root: /home/milosvasic/Projects/
boba
    files scanned: 357
PASS: CM-TEST-MOCK-PID-EXPLICIT-INT clean across 357 file(s)
EXIT=0

$ timeout 60 bash tests/pre_build/
test_bob152_vendored_exclude_scope.sh
...
VERDICT: PASS — BOB-152 exclude is load-bearing (submodules/
excluded, tests/ still caught) for all 3 affected gates.
EXIT=0

git diff --stat scripts/verify-all-constitution-rules.sh after all
mutation/restore cycles above: empty (confirmed byte-identical to
HEAD; no residual mutation left in the working tree, per $11.4.84
quiescence).
```

7. Honest observation (out of scope, NOT fixed here)

Running `covenant_propagation_suite.sh gates --root .` against boba's real, fully-populated tree (submodules initialised) surfaces a large number of BLIND verdicts unrelated to `.worktrees` — e.g. CM-COVENANT-114-208-PROPAGATION: BLIND — carrier-discovery control needle FAILED: 3 of 3 root governance files present on disk were NOT returned by discovery. **I confirmed this is pre-existing and NOT caused by this fix**: it reproduces identically with `config/covenant_propagation_exclusions.tsv` present or temporarily moved aside. It also does not affect the item's acceptance criterion — the hermetic fixture proof (§3/§4) isolates the `.worktrees` behaviour from

this unrelated, larger-scale instability, so the mutation proof is not confounded by it. This looks like a distinct, tracked- worthy defect (possibly related to the engine’s own documented SIGPIPE-class carrier-discovery footgun at scale, or a resource/timing effect under a large real fleet) but is squarely out of BOB-143’s scope and was not touched, investigated further, or fixed here.

8. `pre_build_verification.sh` wiring — left for the conductor

Not needed. `pre_build_verification.sh`’s own CM-COVENANT-PROPAGATION-SUITE invocation (its “Invariant 32”, ADVISORY per its own comment) passes `--root "${PROJECT_ROOT}/constitution"` — the constitution **submodule**’s root — not boba’s repository root, so it was never the source of the `.worktrees`-sourced findings BOB-143 measured and requires no change for this fix. The actual production path that reads `config/covenant_propagation_exclusions.tsv` by default (`scripts/verify-all-constitution-rules.sh`, boba’s own §11.4.32 sweep) needed **no wiring change** either — both the engine and the suite already default to `${root}/config/covenant_propagation_exclusions.tsv`, and that sweep already invokes the suite with `--root "$root"` = boba’s repository root, so simply creating the file was sufficient.

9. Assumptions the item text did not fully specify

1. **Closed exclusion class.** BOB-143’s text does not name which of the engine’s four closed classes (`vendored-third-party` | `generated-code` | `non-shipping-fixtures` | `filename-collision`) applies. I chose `filename-collision`: each orphaned `worktree` is a full historical clone of this repository, so its `CLAUDE.md/AGENTS.md/etc.` share the exact filename the carrier-discovery globs for without being this project’s real, current, owned carriers — the discovery mechanism collides on filename, not on any legitimate governance relationship. This reading is consistent with the class’s own doc comment (“the filename-collision extension”) and with the real-world precedent file I read for format reference (`/home/milosvasic/Projects/vasic/config/ covenant_propagation_exclusions.tsv`, outside this repository).

2. **Item still shows “Queued”**. Per scope, I did not touch docs/workable_items.db or run the workable-items CLI, so BOB-143’s status in the tracker is unchanged by this session; closing/reclassifying it is left to the conductor.
3. The item’s second, already-addressed measurement (cm_test_mock_pid_explicit_int, §1a above) was fixed by an EARLIER, differently-titled commit (c0ea01c) that never updated BOB-143’s own tracker text to say so. I treated this as a pre-existing partial fix to be regression-guarded, not re-done.